

# OSTEOFORGE

MEDTECH

Restore function.
Reconfigure esthetics.
Faster & safer than ever.

**Dr Ramya Bojedla**
Maxillofacial Radiologist (MDS), PhD (IITH)
Founder and CEO

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# Meet The Team



**Dr Ramya Bojedla, MDS, PhD**
*Founder & CEO*

Dr. Ramya Bojedla, Oral Physician and Maxillofacial Radiologist with 4 years of clinical experience, a Ph.D. from IIT Hyderabad, and over 7 years of research expertise, drives innovation in medical device development with her deep clinical insights.



**Dr Falguni Pati, PhD**
*Co-Founder, & Head of Innovation*

With 20 years of experience, Dr. Pati leads research initiatives at OsteoForge, driving innovation and ensuring the company remains at the forefront of biomedical advancements.



**Dr Syam Kumar Y, PhD**
*CSO, Chief Scientific Officer*

Dr. Syam, with over 15 years of expertise in regulatory affairs, ensures OsteoForge's products meet the highest international quality standards and compliance requirements. His expertise guarantees that our innovations adhere to stringent regulatory benchmarks, ensuring global acceptance and trust.



**Dr Aditya Mohan Alwala**
*Clinical Advisor*

Professor, Department of Oral & Maxillofacial Surgery, MNR Dental College & Hospital, Sangareddy.



**Dr Sujit kumar Vakati**
*Clinical Advisor*

MBBS, Ms. Ortho, Mch. Ortho A.A.F, S.E.F (JAPAN), AOT (Germany), IWC (Paris) Dip. In Football Medicine (FIFA)
Consultant Arthroscopy, Joints, Sports Injury Specialist.

Dr. Aditya Mohan Alwala & Dr Sujit kumar Vakati, with over 15 years of clinical experience serves as our Clinical Advisor. His extensive expertise ensures that OsteoForge's solutions are tailored to meet real-world clinical needs, providing practical, patient-centered innovations for craniomaxillofacial and orthopaedic applications.



**Sri Harsha Dagupati**
*Consultant Brand Strategy*

Mr. Sri Harsha Dagupati brings over 20 years of expertise in strategic brand positioning, market visibility, and building brand credibility. His leadership drives OsteoForge's competitive edge, ensuring strong market presence and long-term trust in our innovative solutions

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# The Problem

The urgent need for an alternate solution to metal implants in Craniomaxillofacial and Orthopaedic trauma surgeries.

## Why?

Metal implants in trauma cases have demonstrated the following adverse effects:

**Bone loss** due to stress shielding.

**Restricted bone growth** in pediatric patients.



**MRI relative contraindication**, limiting diagnostic options.

**Additional corrective surgeries**, adding psychological and financial burden on patients.

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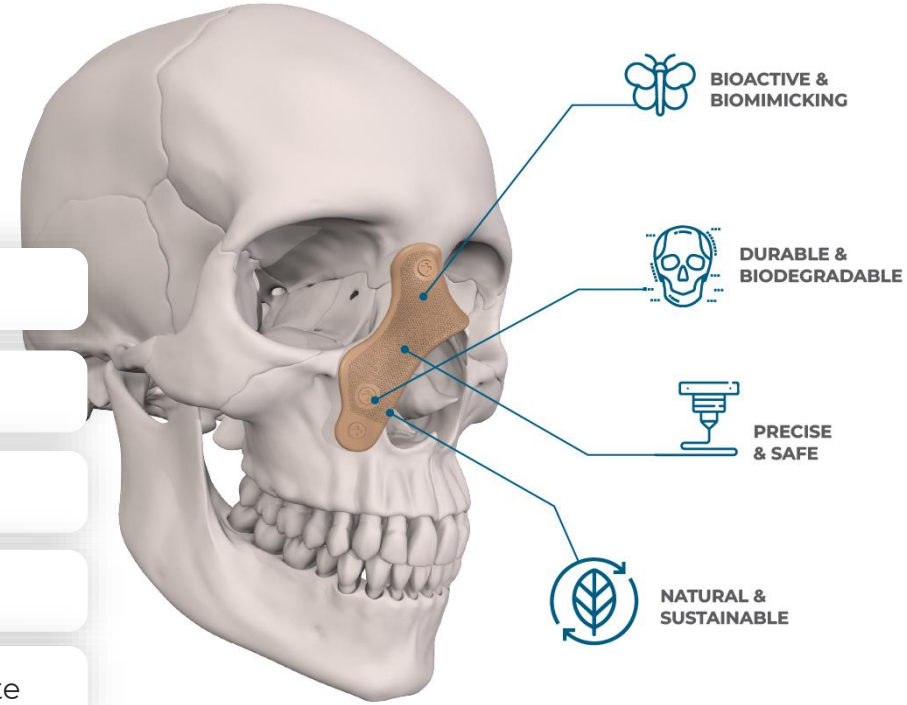
# Seribo

## Our Solution

**First ever**
3D printable silk-based Bioactive and Bioresorbable composite for Craniomaxillofacial and Orthopaedic reconstruction.

**ONE-OF-A-KIND**

- Bone Regenerative
- 3D Printable
- Bioresorbable
- Silk-based Composite



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# Here’s How This Innovation Works

### Bioactive and Biomimetic Scaffolds

- Accelerates healing and enhances bone regeneration (3-fold faster).

### Durable

- Exhibits optimal strength during the entire fracture healing process.

### Intelligent  Bioresorbable and Customizable Scaffolds

- Bioresorbs physiologically in sync with the bone healing process.
- Tailored solutions for both adult and pediatric patients.
- Designed using patient-specific CT data for a perfect fit.



### Natural & Sustainable

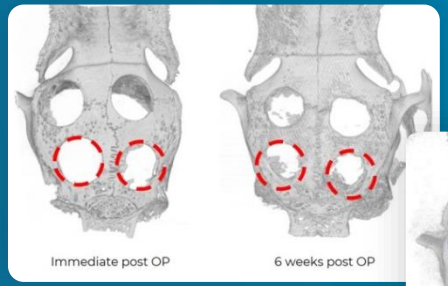
- Crafted from a wild silk variant native to India, reflecting the country’s rich biological and cultural heritage.
- A cornerstone of our sustainable approach, ensuring that innovations contribute to a more responsible and sustainable future.

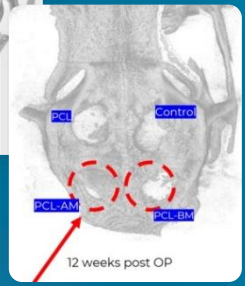
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# Current Traction




**IP India: 202141055556 (2022)**
"Silk Fibroin Microfiber Reinforced Polycaprolactone Composites"




**SERIBO**
2X faster bone regeneration in 12 weeks

**Intellectual Property:**
- ✓ 1 patent granted;
- ✓ 2 patents in the application process.

**Manufacturing & Regulatory:**
- ✓ Leased GMP manufacturing unit.
- ✓ Obtained test batch license from CDSCO.

**Preclinical Testing:**
- ✓ Completed in a GLP-certified lab, confirming safety. (ISO 10993 Biocompatibility)

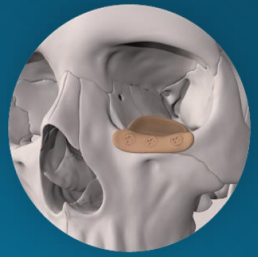
**Clinical Collaboration:**
- ✓ Partnerships with 50 maxillofacial and 7 orthopedic surgeons.
- ✓ Significant interest in 3D-printed Seribo scaffolds.
- ✓ Agreements established with institutions to initiate trials.

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# Product Portfolio – Seribo



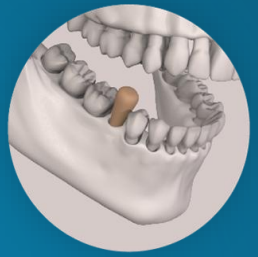
Plates & Screws



Meshes



Scaffolds



Bone Grafts

# B2B Business Model

## TARGET AUDIENCE

- Neurosurgeons
- Orthopedic Surgeons
- Maxillofacial Surgeons
- Esthetic & Plastic Surgeons

## KEY ASSOCIATIONS

- Hospitals
- Institutions
- Trauma Centers
- Tertiary Care Units

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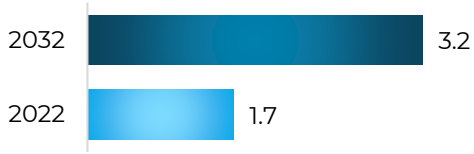
# Bone Plates & Screws Market

## Global Devices Market

**Orthopedic:**
From $60.12B in 2022 to $88.47B in 2032.




**Craniomaxillofacial:**
From $1.7B in 2022 to $3.2B in 2032.



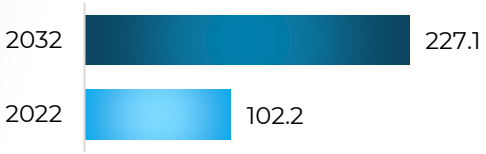



## Indian Devices Market

**Orthopedic:**
From $1.54B in 2022 to $2.4B in 2032.




**Craniomaxillofacial:**
From $102.2M in 2022 to $227.1M in 2032.



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# The Demand

Trauma care needs in India.

**2022 Procedures Data**

## 56.7 Lakhs

Orthopedic fractures
with a **14.2%** annual increase

**Pediatric Orthopedic Fractures**

## 19.3 Lakhs

with an **11%** annual increase

**Craniomaxillofacial Fractures**

## 4.3 Lakhs

with a **7%** annual increase

**Pediatric Craniomaxillofacial Fractures**

## 87,000

with a **7.7%** annual increase



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# BONE GRAFT MARKET

## Global Market

From $2.3 B in 2022

to

$4.3B in 2032.






## Indian Market

From $96.3 M in 2022

to

$390.5 M in 2032.




https://idataresearch.com/product/orthobiologics-market-india/
https://www.alliedmarketresearch.com/bone-graft-substitutes-market#:~:text=Bone%20Grafts%20And%20Substitutes%20Market%20Research%2C%202033,major%20driver%20of%20the%20market

**2022 Procedures Data**

**5.2 Lakhs**

Bone grafting procedures with a **12.4%** annual increase

# Competitive Advantage

**Metal Implants**
**Titanium**

- Bone loss
- **Need** for 2nd Surgery
- Product pricing: 10,000 – 30,000/-

**Polymer Implants**
**(Only International players)**
**PLA based**

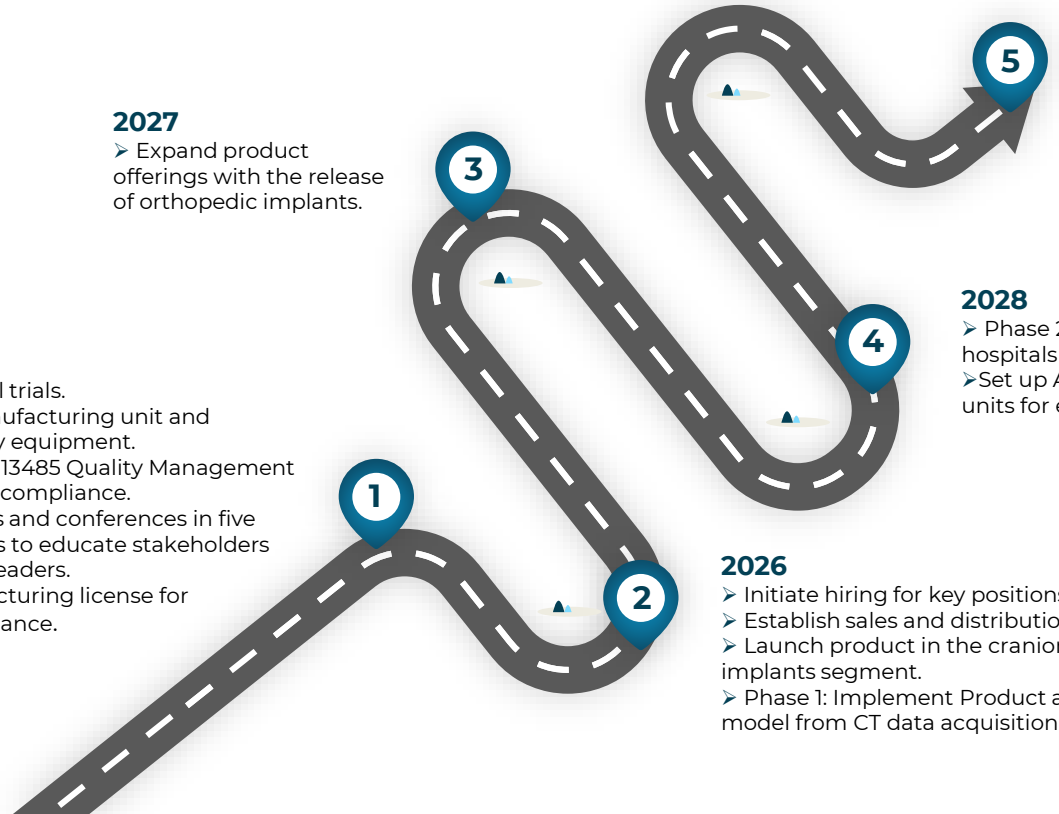
- Reported Risk of infections/ Poor Bone healing
- **NO need** for 2nd Surgery
- Product pricing: 60,000 – 1,20,000/-

stryker
KLS martin GROUP
DePuy Synthes
THE ORTHOPAEDICS COMPANY OF Johnson & Johnson
ZIMMER BIOMET

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- **Enhanced Bone regeneration (~ 2-fold > Faster healing)**
- **Conventional Plates and screws & 3D printed Customized scaffolds**
- **No need for a 2nd Surgery**
- **Indigenous Patented technology**
- **Product pricing: 30,000 – 70,000/-**

# Road Map




**2025**
- Conduct clinical trials.
- Establish a manufacturing unit and procure necessary equipment.
- Implement ISO 13485 Quality Management System (QMS) for compliance.
- Host workshops and conferences in five major metro cities to educate stakeholders and Key opinion leaders.
- Secure manufacturing license for regulatory compliance.

**2026**
- Initiate hiring for key positions to support growth.
- Establish sales and distribution channels.
- Launch product in the craniomaxillofacial implants segment.
- Phase 1: Implement Product as a Service (PAS) model from CT data acquisition to implant delivery.

**2027**
- Expand product offerings with the release of orthopedic implants.

**2028**
- Phase 2: Collaborate with corporate hospitals to broaden market reach.
- Set up AI-enabled in-house 3D printing units for efficient manufacturing.

**2029**
- Obtain USFDA clearance for international product launch.

# Investment Ask

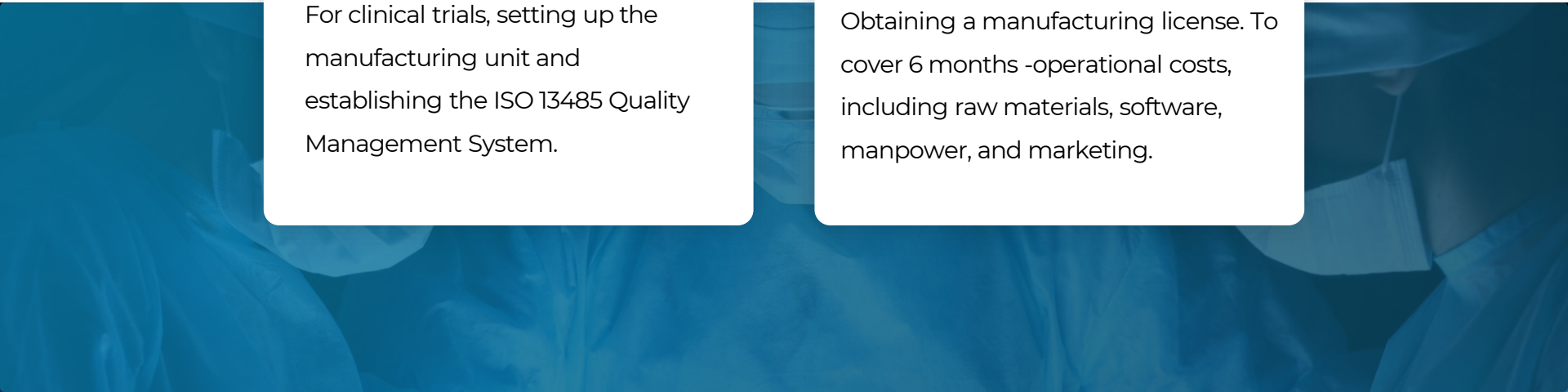
Funding Requirement: **INR 15 Crores**

**INR 8 Crores**

For clinical trials, setting up the manufacturing unit and establishing the ISO 13485 Quality Management System.

**INR 7 Crores**

Obtaining a manufacturing license. To cover 6 months -operational costs, including raw materials, software, manpower, and marketing.



# THANK YOU